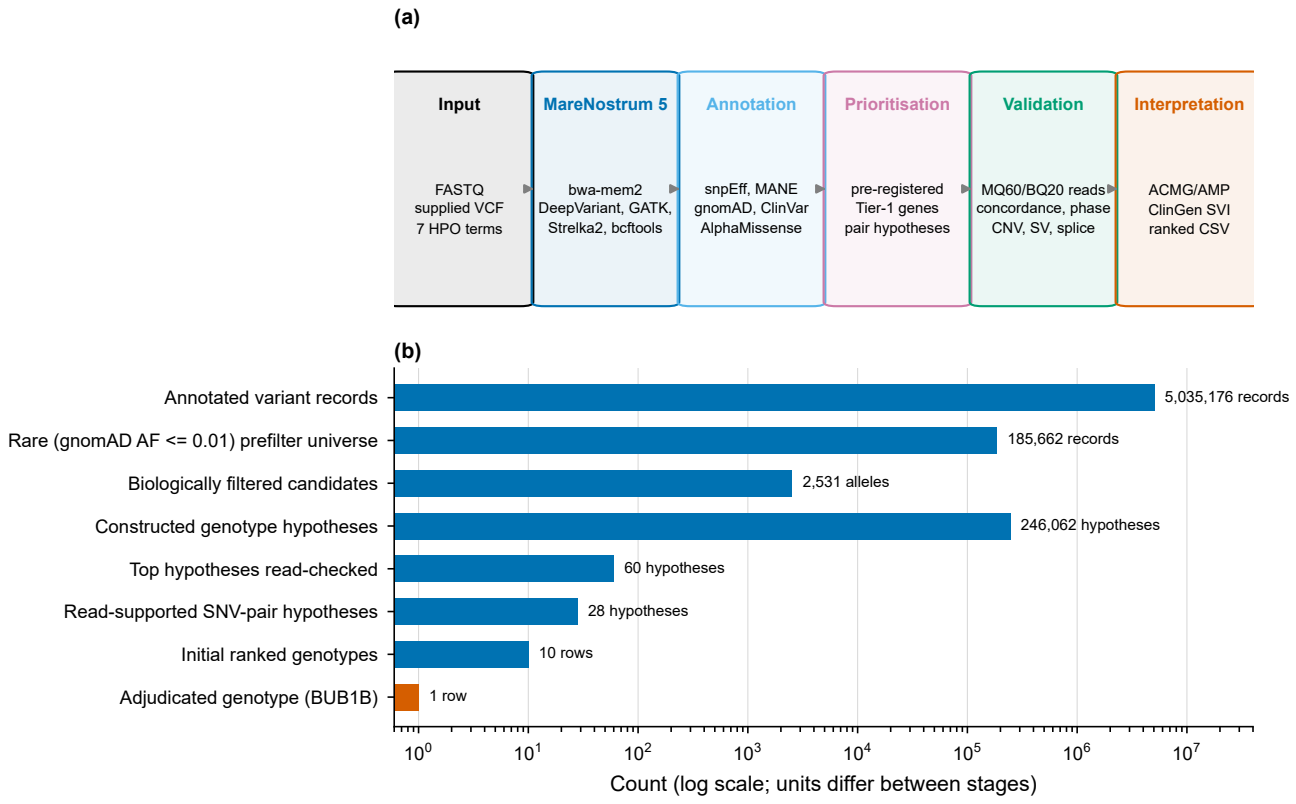
(a)
Input
FASTQ
supplied VCF
7 HPO terms
MareNostrum 5
bwa-mem2
DeepVariant, GATK,
Strelka2, bcftools
Annotation
snpEff, MANE
gnomAD, ClinVar
AlphaMissense
Prioritisation
pre-registered
Tier-1 genes
pair hypotheses
Validation
MQ60/BQ20 reads
concordance, phase
CNV, SV, splice
Interpretation
ACMG/AMP
ClinGen SVI
ranked CSV
(b)
Annotated variant records
Rare (gnomAD AF <= 0.01) prefilter universe
Biologically filtered candidates
Constructed genotype hypotheses
Top hypotheses read-checked
Read-supported SNV-pair hypotheses
Initial ranked genotypes
Adjudicated genotype (BUB1B)
5,035,176 records
185,662 records
2,531 alleles
246,062 hypotheses
60 hypotheses
28 hypotheses
10 rows
1 row
10^0
10^1
10^2
10^3
10^4
10^5
10^6
10^7
Count (log scale; units differ between stages)